

Mynul Islam

Detailed Technical Experience Profile

Software Engineering graduate with hands-on experience building and evaluating NLP classifiers for Bengali, a low-resource language. In my final-year thesis I collected the raw corpus for a five-class emotion dataset, implemented a text CNN in PyTorch, and fine-tuned Bangla BERT, reaching **about 67% accuracy (macro-F1 0.66) on a five-way task**, roughly 20–25 p

oints above my CNN baseline. Multi-class text classification with transformers is the same core technique behind intent detection in chatbots, and I want to apply it to multilingual conversational AI on a Node.js / low-code stack.

PROJECT — EMOTION CLASSIFICATION OF BENGALI SONG LYRICS

An Empirical Approach toward Identifying the Core Sentiments of Bengali Music Using Machine Learning and Neural Network Classifiers

Context	Final-year thesis (course SWE 450), IICT, SUST. Team of three. Submitted February 2024.
Supervisor	Dr. Ahsan Habib, Associate Professor, IICT, SUST
My role	Data Lead & NLP Model Developer — raw data collection, CNN classifier, Bangla-BERT fine-tuning; author of the LaTeX thesis
Tech stack	Python, PyTorch, torchtext, Simple Transformers (Hugging Face), Keras/TensorFlow and scikit-learn (team), LaTeX, Git/GitHub, Google Sheets
Dataset	github.com/Salman067/Dataset_SWE_450 (Data folder)

Business context

Problem. Music platforms tag songs by genre but rarely by emotion, and very little annotated data exists for Bengali. The goal was to classify the dominant emotion of each song verse — Happy, Love, Sad, Depression or Life-oriented — from text alone. We worked at verse level because a single song can move through several emotions.

Applications. Emotion-aware music recommendation, automatic playlist and mood tagging, and mood-aware conversational agents. The underlying building block, multi-class text classification, is the same one used for intent detection in chatbots.

My responsibilities

- **Data collection:** gathered song lyrics from public online sources and organised them by lyricist — 109 lyricists, 325 songs, over 10,000 words — as the raw corpus for verse-level labelling.
- **CNN classifier:** implemented, trained and evaluated a Kim-style text CNN in PyTorch as the neural baseline.
- **BERT classifier:** fine-tuned a Bengali-pretrained BERT with a class-weighted loss and analysed its errors; it was the strongest single neural model in the study.
- **Technical writing:** wrote the 43-page thesis in LaTeX (structure, figures, tables, equations, bibliography).

Team-owned parts (my teammates): verse segmentation and annotation, preprocessing and the rule-based Bengali stemmer, BLSTM, classical ML models (Naive Bayes, SVM, SGD, XGBoost), MLP, and the interactive prediction demo.

Implementation details

A. Dataset and preprocessing (team-built; I supplied the raw data)

- Verse segmentation used two rule sets, chosen per song: blank-line separation, or the Bengali “resting” punctuation mark (danda) that closes a verse.
- Five classes with written annotation rules; mixed-emotion verses were labelled by their dominant sentiment. Annotators agreed on 95% of verses (raw agreement).
- Cleaning: removal of invalid characters and punctuation, Bengali stop-word removal, duplicate removal, and a rule-based Bengali stemmer ported to Python.
- Split: 67.9% train, 12% validation, 20% test.

Happy	Love	Sad	Depression	Life-oriented	Total verses
123	279	324	262	249	1,237

B. CNN classifier (my implementation)

- Architecture follows Kim (2014): embedding layer (100-d, initialised from GloVe 6B) → parallel Conv2d filters with kernel heights 1–5 × embedding size (uni- to 5-gram detectors), 100 filters each → ReLU → max-over-time pooling → concatenation (500 features) → dropout 0.2 → linear layer over 5 classes.
- Adam optimiser, cross-entropy loss, batch size 32 using length-bucketed batches to minimise padding.
- Result: macro precision 0.41, macro recall 0.45. Training loss fell to about 0.6 while validation loss bottomed out near 1.1 around epoch 8 and then rose — a clear overfitting signal on a dataset of about 1.2k verses.

C. BERT classifier (my implementation)

- Fine-tuned *sagorsarker/bangla-bert-base* through Simple Transformers' ClassificationModel with 5 output labels and 3 training epochs.
- Handled class imbalance (123 Happy vs 324 Sad verses) with inverse-frequency class weights, $w = N / (K \times n_{\text{class}})$, passed to the loss.
- Collected wrong predictions per label and inspected a normalised confusion matrix. Confusion concentrated between semantically overlapping classes, for example Life-oriented vs Sad and Happy.
- **Result: accuracy ≈ 0.67; macro precision 0.66, recall 0.67, F1 0.66.** Per-class recall ranged from 0.49 to 0.89.

D. Results across the study (five-class accuracy)

Model	Contributor	Input representation	Result
TF-IDF + Naive Bayes	Team	Count vectors + TF-IDF	0.74
SVM	Team	TF-IDF	0.69
SGD classifier	Team	TF-IDF	0.71
XGBoost	Team	TF-IDF	0.53
MLP	Team	GloVe	0.40
BLSTM	Team	GloVe	0.71
CNN	Me	GloVe-initialised embeddings	P 0.41 / R 0.45 (macro)
Bangla BERT	Me	WordPiece subwords	0.67 (F1 0.66)

In a three-class variant of the task, BERT was the top model (0.65).

What I learned and would do differently

- **Language-specific pretraining matters most.** The Bengali-pretrained BERT beat the CNN by roughly 20–25 points, so for any non-English chatbot the choice of pretrained model matters more than architecture tuning.
- **Small data limits big models.** With about 1.2k verses, TF-IDF + Naive Bayes was competitive with BERT in the five-class setting. Next time I would use stratified k-fold cross-validation and data augmentation for more stable estimates.
- **Embeddings should match the language.** The CNN baseline used English-trained GloVe vectors, which cover Bengali poorly. I would switch to Bengali-native embeddings such as fastText.
- **Label quality sets the ceiling.** Emotions in poetry are subjective. I would report Cohen's kappa rather than raw agreement and add a second annotation pass for ambiguous verses.
- **Next step towards deployment:** wrap the BERT model in a REST API and connect it to a chat interface, which is the natural bridge from this research to conversational AI.

HOW THIS MAPS TO THE ROLE

Requirement in circular	Evidence
Intent-style NLP classification	Five-class text classifier on Bengali, part of a team benchmark of CNN, BLSTM, MLP, BERT and classical baselines; evaluated with precision, recall, F1 and confusion matrices.
Transformers / LLMs	Fine-tuned a pretrained transformer (Bangla BERT) with class-weighted loss and error analysis.
Multilingual conversational flows	Worked with a low-resource, non-Latin-script language: Unicode handling, tokenisation, stemming, stop-word removal and verse segmentation. Comparable challenges arise in Korean, Japanese, Thai and Indonesian flows.
Data quality & testing	Annotation guidelines, agreement checks, de-duplication, and systematic error analysis of wrong predictions.

English communication

43-page English thesis written in LaTeX, plus the final defence presentation.